ISM • SESSION 2 • PROBABILITY & ITS AXIOMS

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The Language of Chance

Sample Spaces, Events, and the Rules They Obey

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Session 1 taught us to describe what we have already seen. But the world is uncertain — the next coin flip, the next customer, the next defective part. Probability is the mathematics of "what *could* happen." This companion builds that language gently: the stage on which chance plays out (the *sample space*), the things we care about (*events*), the few simple rules (*axioms*) every probability must obey, and the two ideas students most often confuse — *mutually exclusive* versus *independent* events.

A friendly companion — read it like a story, not a reference table.

What's inside

1 From "what happened" to "what could happen"

Descriptive statistics (Session 1) looked *backward* at data we already have. Probability looks *forward*, at outcomes that have not happened yet. It is the bridge from describing a sample to reasoning about an uncertain world — the next coin flip, the next customer, the next defective part.

What, why, where — for this whole session

What: a precise language for uncertainty — the stage events play out on (the *sample space*), the things we care about (*events*), and the handful of rules (*axioms*) every probability must obey. **Why:** without rules, "probability" is just opinion; the axioms turn it into mathematics we can compute with and trust. **Where:** this vocabulary underlies *everything* ahead — conditional probability, Bayes' theorem, every distribution, and every model that reasons under uncertainty.

Definition 1.1: Random experiment

A **random experiment** is any action whose outcome cannot be known in advance — even if we repeat it under identical conditions.

You run random experiments all day

Flipping a coin (heads or tails?), waiting for a bus (how many minutes?), giving a lecture (how many students show up?), sending a signal down a noisy channel (what arrives at the far end?). In each case the *rules* are fixed but the *result* is not — that uncertainty is precisely what "random" means. Probability is how we put numbers on that uncertainty.

2 The three nested ideas: sample space, outcome, event

To reason about chance we need three ideas that fit inside one another like nesting dolls: the full set of possibilities, a single result, and a collection of results we care about.

Definition 2.1: Sample space, outcome, event

- The **sample space** S is the set of *all* possible outcomes. Rolling a die: $S = \{1, 2, 3, 4, 5, 6\}$. Testing a chip: $S = \{\text{accept}, \text{reject}\}$.
- An **outcome** is one element of S — a single result.
- An **event** is any *subset* of S — a bundle of outcomes we care about, such as "the die shows an even number" $= \{2, 4, 6\}$.

Each time we run the experiment, an event A either *occurs* (the outcome lands inside A) or *does not* (it lands outside).

A theatre stage

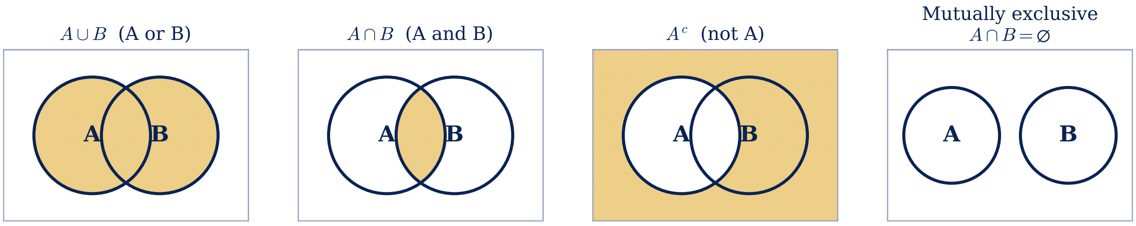
Picture the sample space S as a whole stage — every spot an actor could stand on. An *outcome* is one exact spot. An *event* is a chalk-marked region of the stage; the event "occurs" if the actor happens to land inside that region. Two special regions always exist: the *empty* region $\varnothing$ (impossible — the actor is off-stage) and the *whole* stage S (certain — the actor is somewhere).

Sample spaces can be tiny or enormous

A coin has $|S| = 2$; a die has 6; two dice have 36; monitoring three phone calls as voice/data has $2^3 = 8$. Sometimes S is even infinite (the *exact* minutes you wait for a bus). Knowing the size and shape of S is the first move in almost every probability problem.

3 Building new events from old ones

Because events are just sets, the set operations give them meaning — and Venn diagrams let us *see* them.



The four building blocks. ***Union*** *$A \cup B$ shades everything in A or B (or both).* ***Intersection*** *$A \cap B$ shades only the overlap — A and B together.* ***Complement*** *A^c shades everything* not *in A.* ***Mutually exclusive*** *events do not overlap at all.*

- **Union** $A \cup B$ — "*A or B* (or both) happens."
- **Intersection** $A \cap B$ — "*A and B* both happen."
- **Complement** A^c — "A does *not* happen."

The complement trick — a genuine time-saver

Since A either happens or it doesn't, $P(A) + P(A^c) = 1$, so $P(A) = 1 - P(A^c)$. Whenever a question contains the words "at least one," the *opposite* ("none") is usually far easier to count. Computing $1 - P(\text{none})$ then hands you the answer almost for free — a move we will use again and again.

4 The classic confusion: mutually exclusive vs. independent

These two ideas trip up nearly every beginner, because both sound like "the events are unrelated." They are in fact almost *opposite*. Let us settle it for good.

Definition 4.1: Mutually exclusive vs. independent

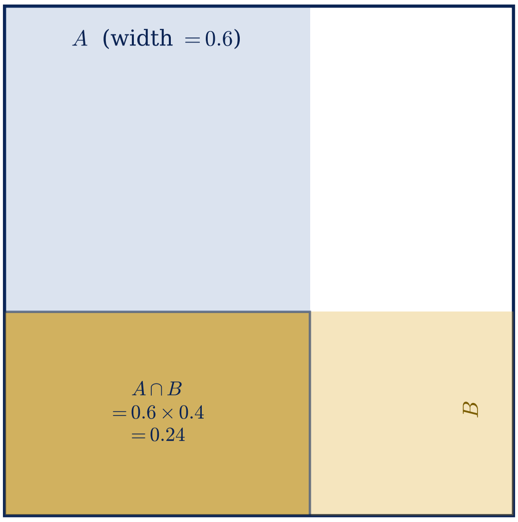
- **Mutually exclusive** (disjoint): the events *cannot happen together*, $A \cap B = \varnothing$, so $P(A \cap B) = 0$. One happening *rules out* the other.
- **Independent**: one happening gives *no information* about the other, $P(A \cap B) = P(A)\,P(B)$.

Two switches, two stories

Mutually exclusive is a single light switch: it is either *up* or *down*, never both — choosing one forbids the other. **Independent** is two switches in different rooms: flipping one tells you nothing about the other. "Can't happen together" and "don't affect each other" are completely different statements.

	Mutually exclusive	**Independent**
Key equation	$P(A \cap B) = 0$	$P(A \cap B) = P(A)P(B)$
Plain meaning	can't both happen	don't influence each other
Knowing A happened...	tells you B *didn't*	tells you *nothing* about B
Everyday example	heads vs. tails on *one* flip	heads on flip 1 vs. flip 2

Independence = the overlap is exactly
the product of the two strips



A picture of independence. If A occupies a fraction 0.6 *of the width and B a fraction* 0.4 *of the height, then* — when they don't influence each other — *their overlap is the rectangle of area* $0.6 \times 0.4 = 0.24$. *Independence is precisely the statement "*$P(A \cap B) = P(A)\,P(B)$*."*

They can't be both (when probabilities are positive)

If two events with positive probability are mutually exclusive, then one happening tells you the other *definitely didn't* — that is the strongest possible information, the exact opposite of "no information." So such events can *never* be independent. Mixing these up is the single most common early-probability error.

Worked example 4.1: Testing independence with two dice

Throw a pair of dice and define $A = \{(1,2),(2,1),(2,2)\}$ and $B = \{(2,2),(2,3),(2,4),(2,5),(2,6),(3,2),(4,2),(5,2),(6,2)\}$. Then

$$P(A) = \tfrac{3}{36}, \quad P(B) = \tfrac{9}{36}, \quad P(A \cap B) = P\{(2,2)\} = \tfrac{1}{36}.$$

The independence test asks: does $P(A \cap B) = P(A)P(B)$? Here $P(A)P(B) = \frac{3}{36} \cdot \frac{9}{36} = \frac{27}{1296} = \frac{1}{48}$, which is *not* $\frac{1}{36}$. Since the two sides differ, A and B are **dependent** — knowing one shifts the odds of the other.

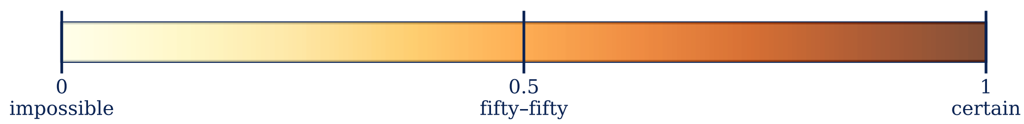
5 What exactly *is* a probability?

There are three doorways to a probability, and they agree when the setup is fair.

Three doorways to the same number

- **Classical** (count, when outcomes are equally likely): $P(A) = \dfrac{\text{favourable outcomes}}{\text{total outcomes}}$. A fair die: $P(\text{even}) = \frac{3}{6}$.
- **Empirical** (measure by repeating): $P(A) \approx \dfrac{\text{times } A \text{ occurred}}{\text{number of trials}}$. Flip a coin 10,000 times and the fraction of heads hugs 0.5.
- **Axiomatic** (the modern, rigorous route): forget where the number came from; just insist it obeys the rules below.

Probability lives on a scale from 0 to 1



Every probability lives between 0 (impossible) and 1 (certain). A value of 0.5 means the event is exactly as likely to happen as not.

Key formula 5.1: The axioms of probability

A probability P assigns each event a number obeying:

1. **Non-negativity:** $P(A) \geq 0$.
2. **Certainty:** $P(S) = 1$ — something must happen.
3. **Additivity:** if A and B are mutually exclusive, $P(A \cup B) = P(A) + P(B)$.

Everything else in probability is built from just these three.

Why these three rules are pure common sense

(1) You can't have a -30% chance of rain. **(2)** *Some* outcome must occur, so "any outcome at all" has chance 100%. **(3)** If two things can't co-occur, the chance of "one or the other" is simply their chances added. From these alone, two everyday facts pop out for free: $P(A^c) = 1 - P(A)$ and $P(\varnothing) = 0$.

Worked example 5.1: Is this a legal set of probabilities?

Four mutually exclusive outcomes A, B, C, D are each assigned a probability. A list is *permissible* only if every value is ≥ 0 **and** they sum to 1.

$$\begin{aligned}&\text{(a) } 0.38, 0.16, 0.11, 0.35: \quad \text{sum} = 1.00, \text{ all} \geq 0 \ \checkmark\\ &\text{(b) } 0.31, 0.27, 0.28, 0.16: \quad \text{sum} = 1.02 \ \text{✗}\\ &\text{(c) one value is } -0.06 < 0 \ \text{✗}\\ &\text{(d) } \tfrac{1}{2}, \tfrac{1}{4}, \tfrac{1}{8}, \tfrac{1}{16}: \quad \text{sum} = \tfrac{15}{16} \neq 1 \ \text{✗}\end{aligned}$$

Only (a) is permissible. Two quick checks — "no negatives" and "adds to one" — catch every illegal assignment.

6 The addition rule: don't double-count the overlap

How likely is "A or B"? Adding $P(A) + P(B)$ counts the overlap $A \cap B$ *twice*, so we subtract it back once.

Key formula 6.1: General addition rule

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

If A and B are mutually exclusive, the overlap is empty and this collapses to $P(A \cup B) = P(A) + P(B)$.

Counting a class without counting twice

30 students take maths and 20 take chemistry, but 10 take *both*. How many take at least one subject? Not $30 + 20 = 50$ — that counts the 10 double-enrolled students twice. The honest answer is $30 + 20 - 10 = 40$. The "$- P(A \cap B)$" is exactly that correction for double-counting.

Worked example 6.1: Passing the exams

A student passes statistics with $P(S) = \frac{2}{3}$ and maths with $P(M) = 1 - \frac{5}{9} = \frac{4}{9}$, and passes *at least one* with $P(S \cup M) = \frac{4}{5}$. The chance of passing *both* comes from rearranging the addition rule:

$$P(S \cap M) = P(S) + P(M) - P(S \cup M) = \tfrac{2}{3} + \tfrac{4}{9} - \tfrac{4}{5} = \tfrac{14}{45}.$$

Worked example 6.2: Investors in both markets

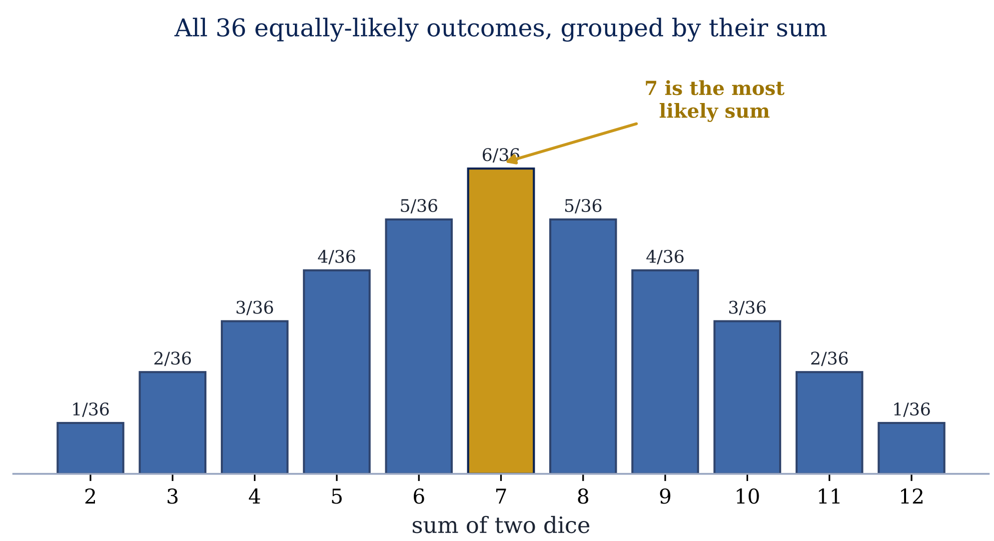
75% invest in annuities (A), 45% in the stock market (B), and 85% in *at least one*. What fraction invest in *both*? Again solve the addition rule for the overlap:

$$P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.75 + 0.45 - 0.85 = 0.35.$$

So 35% are in both — and notice A and B here are clearly *not* mutually exclusive, since their overlap is non-zero.

7 Counting your way to a probability

When every outcome is equally likely, probability becomes careful *counting*: favourable outcomes over total outcomes.



Two fair dice give 36 equally-likely outcomes. Grouped by their sum, the counts form a triangle peaking at 7 — the sum with the most ways to occur.

Worked example 7.1: Two dice: three quick questions

Using the 36 outcomes above:

$$P(\text{sum} > 8) = \frac{4+3+2+1}{36} = \frac{10}{36} = \frac{5}{18}, \qquad P(\text{sum} < 6) = \frac{1+2+3+4}{36} = \frac{10}{36} = \frac{5}{18}.$$

For "the sum is *neither* 7 nor 11," the complement trick shines: sums of 7 or 11 occur in $6 + 2 = 8$ ways, so

$$P(\text{neither 7 nor 11}) = 1 - \frac{8}{36} = \frac{28}{36} = \frac{7}{9}.$$

Worked example 7.2: A committee with a majority of women (counting with choices)

Choose 5 people from 8 men and 4 women. A women-majority means 4 women+1 man, or 3 women+2 men. Counting the ways with combinations $\binom{n}{k}$ ("n choose k"):

$$P = \frac{\binom{8}{1}\binom{4}{4} + \binom{8}{2}\binom{4}{3}}{\binom{12}{5}} = \frac{8+112}{792} = \frac{120}{792} = \frac{5}{33}.$$

The recipe is always the same: count the favourable arrangements, divide by all arrangements.

8 Practice problems

Your turn

1. (complement trick). A bank's accounts are 40% savings, 35% current, the rest loans. Find $P(\text{loan})$, $P(\text{not savings})$, and $P(\text{current or loan})$.

2. (addition rule + Venn). In a suburb, 60% of homes have internet, 80% have TV, and 50% have both. What fraction have *at least one* service? *Exactly one*? Draw the two-circle Venn first — the picture makes the answer obvious.

3. (contradiction). A tells the truth 80% of the time, B 60%. In what fraction of cases do they *contradict* each other on the same fact? *(Hint: they contradict when exactly one of them lies.)*

4. (max of a product). If A and B are mutually exclusive with $A \cup B = S$, what is the largest possible value of $P(A)\,P(B)$? *(Hint: $P(A) + P(B) = 1$; a product with a fixed sum is biggest when the two parts are equal.)*

The thread to the next session

So far every probability has described the world "in general." But real reasoning is *conditional*: *given* a positive test, how likely is the disease? *Given* that it rained, how likely is the job to finish on time? Updating a probability in the light of fresh information is the heart of Session 3 — **conditional and total probability**.